

Forensic Comparison of Soil Particles Using Gaussian Mixture Models and Likelihood Ratio Test

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Abstract—Forensic analysis of soil traces can be highly valuable in criminal investigation as it can provide evidence which links a person or a contaminated object with one specific location. Given two soil samples, the task of a forensic expert is typically to decide whether they both originate from the same location or not. To confidently answer this question it is necessary to perform a complex analysis focused on examination of the sample’s organic, anthropogenic, and naturally occurring components. In this paper, we focus on one element of the analysis which studies small mineral particles within the sample. In particular, we propose a novel method for automatic comparison of soil particles, using scanning electron microscope images acquired in the secondary electron (SE) or backscattered electron mode. The method involves segmentation of particles, identification of their contours, and extraction of local descriptors from the images, which are then used to train a sample specific and non-specific Gaussian mixture model (GMM). Finally, a likelihood ratio, based on the GMMs, is calculated to assess the odds that two samples originate from the same location. The proposed method, utilizing Root SIFT descriptors extracted from the SE images along the particle contours, achieved an *equal error rate* of 13.1 % and an *area under the curve* of 95.2 %, surpassing our baseline method derived from particle size analysis.

Keywords—soil analysis, scanning electron microscopy, deep learning, segmentation, Gaussian mixture model

I. INTRODUCTION

Forensic science plays a crucial role in criminal investigations, providing valuable evidence to unravel criminal acts. The analysis of soil traces that have adhered to clothing, shoes, vehicles, or other objects is often required in forensic expert practice to confirm the place of origin of contamination [1]. In accomplishing these goals, the forensic expert is typically faced with the problem of deciding whether two soil samples are comparable and, if so, whether they originate from the same location. As per [2], this decision can be made with high confidence from a detailed analysis of the mineralogical and chemical characteristics of the samples. The standard procedure of forensic soil analysis typically involves qualitative characterization of macroscopic soil morphology, color, consistency, etc., followed by separate analysis of organic and mineral components of the sample [1]. Analysis of the mineral components involves several steps that process the soil as a whole. After that, it is, nevertheless, common practice to separate the mineral particles of the sample from clay and other components by washing. The separated mineral particles are then subjected to microscopic analysis, which allows characterization of the *morphological parameters* of

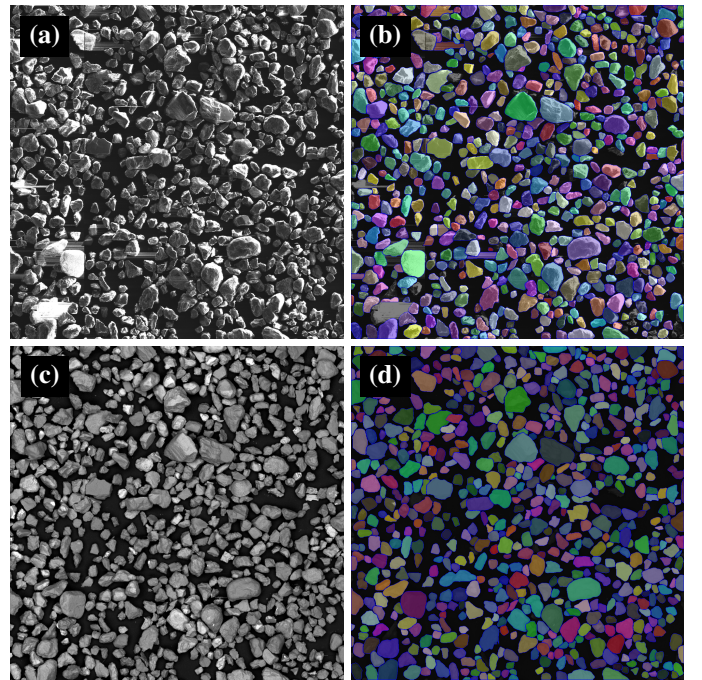


Fig. 1: Sample images of the natural mineral part of soil acquired by a scanning electron microscope in the (a) SE mode, (c) BSE mode, and their corresponding segmentation masks (b), (d). Note that segmentation is an intermediate result of the proposed method.

the particle population, as well as its *elemental* (chemical) *composition*. Possible morphological parameters include particle equivalent diameters, perimeter, area, roundness, convexity, or particle texture. A high significance has been attributed especially to *particle size analysis*, which characterizes soil samples on the basis of particle size distribution [3]. Elemental analysis, on the other hand, characterizes the soil samples on the basis of individual chemical elements, or minerals. Frequently used techniques for this type of analysis include a scanning electron microscope (SEM) equipped with an energy-dispersive spectrometer (EDS).

SEM allows both characterization of morphological parameters and elemental composition of the analyzed sample. In SEM, the interaction of an electron beam with the sample atoms generates various types of measurable signals. In the secondary electron (SE) and backscattered electron (BSE) mode, the SEM measures the energy of the received secondary

and backscattered electrons, respectively, which are then visualized in a grayscale image [4, Sec. 8.3]. The gray value of the BSE image has a strong relation to the atomic number of the observed elements; it is, therefore, useful in discriminating elements with distinct atomic numbers. In the EDS mode, the SEM is equipped with a spectrometer that measures X-ray energy emitted during the interaction. Peaks in the resulting energy spectrum can then be used to differentiate elements even if their atomic numbers are similar.

Image processing, and statistical analysis techniques, can be used to create automated methods for soil sample analysis. In [5] the authors proposed an automated method for quartz grain texture analysis which can quickly exclude grains that are dissimilar to a control sample. This alleviates the manual labor of the forensic expert, who would otherwise have to analyze all the grains manually. In [6] the authors proposed an automated method for assessment of soil sample similarity based on elemental analysis. Using measurements from ED-XRF spectrometer they build a distribution of 20 elements, which are then compared using Canberra distance. To the best of our knowledge, no existing literature considers the use of advanced image processing techniques to classify soil samples from SEM images.

Forensic comparison of soil samples is in a sense similar to the task of authorship verification from handwritten text [7], [8]. In authorship verification, we are given a disputable and a reference sample of handwriting, and our goal is to decide whether the two samples are comparable and, if so, whether they share a common author. The state-of-the-art methods for handwriting verification are surprisingly general, in that they can be applied to almost any kind of images. In [7], feature vectors based on texture descriptors, such as local binary patterns and local phase quantization, are extracted from preprocessed handwriting images. The feature vectors of disputable and reference samples are then classified as “same author” or “different author” using a support vector machine (SVM) trained on a large quantity of labeled samples. In [9] and [10], scale-invariant feature transform (SIFT) descriptors are calculated along the contours of individual letters, which are then combined into an aggregate feature vector using a MAP-adapted Gaussian mixture model (GMM). The feature vectors are then again classified using an SVM.

As we will discuss in Section IV the contours of individual particles carry important information for discriminating sample similarity. Nevertheless, extraction of the particle contours, which is equivalent to the image segmentation, is a non-trivial problem. Several classical approaches such as threshold-based, edge-based, and region-based segmentation [11] might be suitable for segmentation of clearly separable particles, they are limited in handling of the complex image structures where particles are in contact with each other, or one particle covers another particle. We expect that much better results can be obtained with the use of deep learning methods. Convolutional neural networks have been shown to outperform classical segmentation methods by taking advantage of large amounts of available data for training [12]. However, this strength is also a limitation as these techniques need a large corpus of training data, which is not always available. The Segment Anything

Model (SAM) [13] is able to provide precise segmentation without the need of extensive training data. Its unique zero-shot capability facilitates transfer to new image modalities and tasks without the need for additional training.

In this paper, we develop a novel method for automated comparison of soil samples, based on SEM images of the soil particles, acquired in the SE or BSE mode. In our method, we take inspiration from the methods [9], [10] for authorship verification from handwritten text. First, using SAM, we segment individual mineral particles in order to extract their contours. Then we calculate diverse local descriptors along the contours, which are used to train a GMM model. In agreement with the European Network of Forensic Science Institutes (ENFSI) guidelines for reporting evaluative evidence [14], we then use the GMM model to calculate the odds (or posterior odds) that two given soil samples originate from the same location. If needed, the odds can be thresholded to obtain a hard decision, or, more practically, it can be combined with the result of a standard analysis.

II. MATERIALS AND METHODS

A. Soil samples

The Institute of Criminalistics in Prague provided us with 30 SEM images (SE and BSE mode) of soil samples collected at different locations in the Czech Republic. The acquisition of the images involved the following chain of operations, which were performed by an expert. First, large organic or anthropogenic particles such as roots, leaves, or pieces of brick or glass were removed from the samples by hand. Then the samples were washed to remove any clay and organic components. The resulting particles were then rinsed in ethanol, dried out, and transferred onto a conductive plate. The prepared sample was then coated with a thin layer of carbon and analyzed using a high-resolution SEM (Tima 3 GMU, Tescan, Brno, Czech Republic) equipped with a secondary electron and backscattered electron detector. Examples of the acquired grayscale images are shown in Fig. 1.

B. Proposed method

The proposed method involves a *training stage* and a *testing stage*, both of which share a common preprocessing pipeline. This pipeline proceeds as follows. First, we segment individual soil mineral particles in the input SEM images, obtaining a set of particle contours. The neighborhood of these contours is expected to carry discriminative information about the sample, such as particle shape and texture. Along the contours, we spread keypoints, which identify positions in the image where local descriptors will be calculated. This is achieved by placing a keypoint at the position of every k -th pixel along the contour. At each keypoint position, we calculate a descriptor based on pixels in its local neighborhood. In the training stage, a large set of local descriptors extracted from a training dataset of SEM images is used to train a background GMM. In the testing stage, we are given SEM images of a *control* and *questioned* soil samples, whose descriptors, together with the trained background GMM, are used to calculate the odds that the two samples originate from the same location. Details of the individual steps are described in the following.

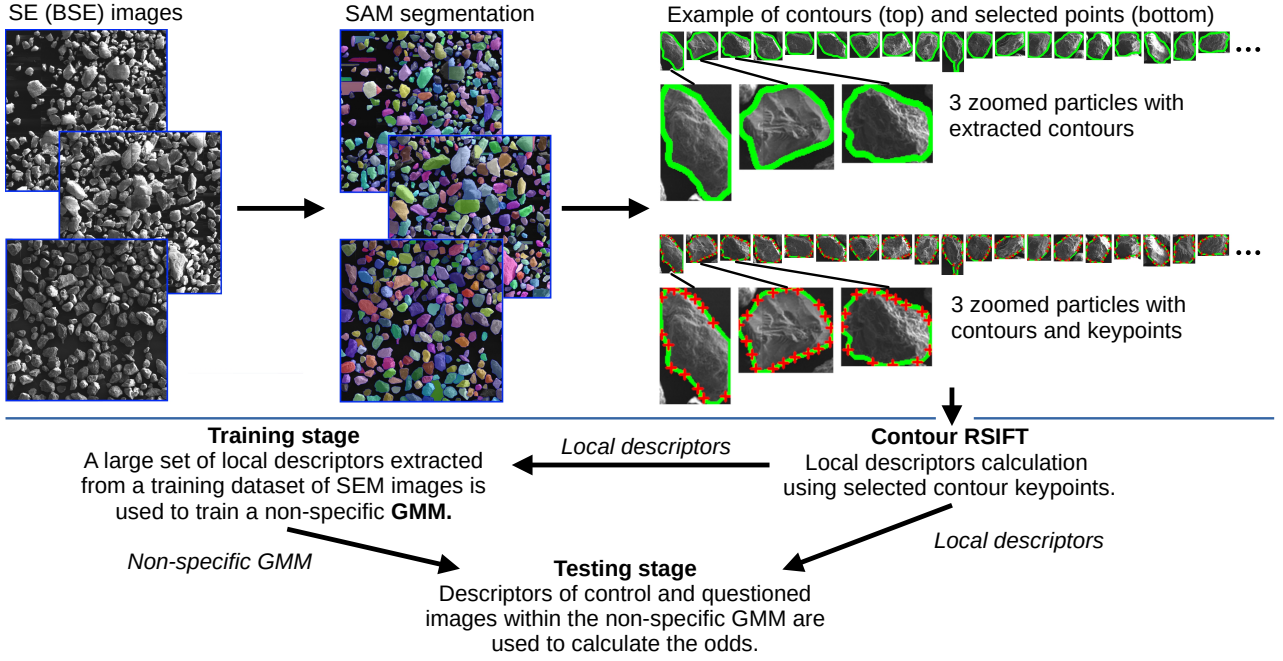


Fig. 2: Schematic illustration of the proposed method.

a) Segmentation: Each grayscale SE or BSE image in a resolution of 1024×1024 pixels is used as input to SAM which generates label maps using a regular 64×64 point grid as a prompt. When the label maps are obtained, contours are created via standard OpenCV functions [15].

b) Descriptor extraction: At each keypoint position, we calculate Root SIFT (RSIFT) descriptors [16], which we found to perform better than standard SIFT. RSIFT descriptors are calculated by first L1 normalizing the standard SIFT descriptor and then taking the square root of each element. Additional parameters required to calculate the SIFT descriptors include the keypoint size, determining the extent of the local neighborhood used in the calculation, and the keypoint orientation. In all experiments, we set the keypoint size to 11px. We tested aligning the keypoint orientation with the particle contour. However, better performance was achieved by setting the orientation to 0° . Finally, we compress the 128-dimensional descriptor to 64 dimensions using principal component analysis (PCA). The parameters of the PCA transformation are learned on a training dataset.

c) GMM training and adaptation: In this paper, we use GMMs as flexible parametric approximations of the distributions of image-based descriptors. For the problem at hand, we need to probe the distribution of all possible descriptors that can appear under natural conditions, which we approximate by a “non-specific” GMM, and the distribution of descriptors for one given soil sample, which we approximate by a “specific” GMM. Denoting the descriptor as $\mathbf{x}$ the general formula for the probability density function (pdf) of a GMM is given by $p(\mathbf{x}; \Theta) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, which is parametrized by $\Theta \triangleq \{\pi_1, \dots, \pi_K, \boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K, \boldsymbol{\Sigma}_1, \dots, \boldsymbol{\Sigma}_K\}$. In particular, for the non-specific GMM we use a number of descriptors from the training dataset to determine the parameters by standard expectation-maximization (EM) algorithm, maximizing

the log-likelihood [17]. The specific GMM, on the other hand, is obtained by adapting the parameters of the non-specific GMM using the following two-step method [18, Sec. 3.4]. Let $\mathcal{X}_c \triangleq \{\mathbf{x}_c^{(1)}, \dots, \mathbf{x}_c^{(N)}\}$ denote N descriptors originating from the control soil sample. Let $\Theta = \{\pi_1, \dots, \pi_K, \boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K, \boldsymbol{\Sigma}_1, \dots, \boldsymbol{\Sigma}_K\}$, denote the weights, mean vectors, and covariance matrices of the non-specific GMM. In the first step, we calculate the probability that the n -th descriptor was drawn from the k -th component,

$$P(k, n) = \frac{\pi_k \mathcal{N}(\mathbf{x}_c^{(n)}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_c^{(n)}; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}, \quad (1)$$

and the sufficient statistics of the weight, mean, and variance parameters

$$c_k \triangleq \sum_{n=1}^N P(k, n), \quad (2)$$

$$\mathbf{m}_k \triangleq \frac{1}{c_k} \sum_{n=1}^N P(k, n) \mathbf{x}_c^{(n)}, \quad (3)$$

$$\mathbf{S}_k \triangleq \frac{1}{c_k} \sum_{n=1}^N P(k, n) \mathbf{x}_c^{(n)} \mathbf{x}_c^{(n)T}, \quad (4)$$

for $k = 1, 2, \dots, K$. In the second step, sufficient statistics are used to update the parameters of the non-specific GMM using

$$\pi'_k \propto \frac{\alpha_k c_k}{K} + (1 - \alpha_k) \pi_k, \quad (5)$$

$$\boldsymbol{\mu}'_k = \alpha_k \mathbf{m}_k + (1 - \alpha_k) \boldsymbol{\mu}_k, \quad (6)$$

$$\boldsymbol{\Sigma}'_k = \alpha_k \mathbf{S}_k + (1 - \alpha_k) (\boldsymbol{\Sigma}_k + \boldsymbol{\mu}_k \boldsymbol{\mu}_k^T) - \boldsymbol{\mu}'_k \boldsymbol{\mu}'_k^T, \quad (7)$$

for $k = 1, \dots, K$, where $\alpha_k \triangleq \frac{\pi_k}{\pi_k + r}$ and $r > 0$ is a fixed “relevance factor” parameter. The proportionality factor for π'_k is chosen such that $\sum_{k=1}^K \pi'_k = 1$. As a result we obtain adapted parameters $\Theta' = \{\pi'_1, \dots, \pi'_K, \boldsymbol{\mu}'_1, \dots, \boldsymbol{\mu}'_K, \boldsymbol{\Sigma}'_1, \dots, \boldsymbol{\Sigma}'_K\}$.

d) *Likelihood ratio calculation*: To calculate the odds that a questioned sample comes from the same location as the control one, we consider two alternative hypotheses. Hypothesis H_0 states that the questioned sample originates from the same location as the control sample. On the contrary, hypothesis H_1 states that the questioned sample originates from a different location than the control sample. The two hypotheses are characterized by the conditional pdfs $p(\mathbf{x}|H_0)$ and $p(\mathbf{x}|H_1)$. In this work, the two conditional pdfs are given by the specific and non-specific Gaussian mixtures, i.e., $p(\mathbf{x}|H_0) \triangleq p(\mathbf{x}|\Theta')$, $p(\mathbf{x}|H_1) \triangleq p(\mathbf{x}|\Theta)$, where Θ denotes the parameters of the non-specific model calculated by the EM algorithm, and Θ' denotes the adapted parameters calculated via (5)–(7). Given N descriptors $\mathcal{X}_q = \{\mathbf{x}_q^{(1)}, \dots, \mathbf{x}_q^{(N)}\}$ originating from a questioned soil sample the odds in favor of H_0 is calculated as

$$\mathcal{L}_0 \triangleq \frac{p(\mathcal{X}_q|H_0)}{p(\mathcal{X}_q|H_1)} = \prod_{n=1}^N \frac{p(\mathbf{x}_q^{(n)}|H_0)}{p(\mathbf{x}_q^{(n)}|H_1)}, \quad (8)$$

where we assumed that $\mathbf{x}_q^{(n)}$, for $n = 1, \dots, N$, are conditionally independent and identically distributed given the hypothesis. Additionally, if we have access the prior odds $\mathcal{P}_0 \triangleq \frac{p(H_0)}{p(H_1)}$, which represent our prior belief that H_0 is true (perhaps based on other evidence), we can combine it with the result of the current analysis by calculating the posterior odds [19] $\mathcal{L}'_0 = \mathcal{L}_0 \cdot \mathcal{P}_0$. Finally, to obtain a hard decision, we compare $\mathcal{L}_0$ or $\mathcal{L}'_0$ with a chosen threshold $\gamma \in \mathbb{R}$ and accept hypothesis H_0 whenever it is larger than γ , or hypothesis H_1 whenever it is smaller than γ .

III. EXPERIMENTS

As already discussed in Section II-B, the preprocessing pipeline of the proposed method involves segmentation of the soil mineral particles followed by extraction of RSIFT descriptors along the particle contours; we will refer to this implementation as “Contour RSIFT”. In our experiments, we additionally consider two modifications of the preprocessing pipeline and an additional baseline method. The first modification denoted simply as “RSIFT”, replaces the segmentation step by the standard SIFT detector [20]. The second modification, denoted as “Morphology”, replaces RSIFT by calculation of several morphological parameters of the mineral particles, such as equivalent area/perimeter diameters, Feret min/max diameter, and many others. In particular, we use all the morphological parameters that are available in the *imea* package [21]. Finally, our baseline method, denoted as “Particle size”, derives from particle size analysis. In this method, we calculate a histogram of particle sizes for the control and questioned samples, and then we compare the two histograms using the Hellinger distance [16]. Whenever the Hellinger distance is smaller than a chosen threshold, we accept H_0 , otherwise, we accept H_1 . As a measure of particle size, we use the equivalent area diameter.

A. Dataset

The dataset of 30 SEM images (see Section II-A for details about the image acquisition) was split into a training subset of 24 images and a testing subset of 6 images. All images

were furthermore cut into four equally sized square patches producing a training dataset of 96 images and testing dataset of 24 images. We stress that cutting of the images into four patches is essential, because it generates *positive* pairs, i.e., pairs of images that originate from the same location. Positive pairs together with *negative* pairs, i.e., pairs of images that originate from a different location, are required for calculation of the performance metrics that we describe in the following section. In total the testing dataset contains $\binom{24}{2} = 276$ pairs, $3 \cdot 4 \cdot 16 = 72$ of them are positive and $276 - 72 = 204$ are negative. Note that the number of positive/negative pairs in the training dataset is irrelevant because GMM training, in contrast to evaluation, does not process the images in pairs (see Section II-B for details on GMM training).

B. Performance metrics

To evaluate the performance of our method, we iterate over all 276 pairs of the testing dataset, and for each pair, we calculate $\mathcal{L}_0$, treating the two items as control and questioned samples. Over the whole dataset, we count the number of false positives, i.e., $\mathcal{L}_0 > \gamma$ for a negative pair, and false negatives, i.e., $\mathcal{L}_0 < \gamma$ for a positive pair. Dividing false positives and false negatives by the total number of positive and negative samples, respectively, we obtain the false positive rate and the false negative rate. The *equal error rate* (EER) is then defined as the false positive rate (or equivalently, the false negative rate) for γ chosen such that the two rates are (approximately) equal. The *area under curve* (AUC) is defined as the area under the receiver operating characteristic.

IV. RESULTS AND DISCUSSION

Performance metrics for the proposed and reference methods evaluated on the testing dataset (SE mode and BSE mode) are reported in Table I. We see that the baseline method (Particle size) performs surprisingly well on the considered data. With EER of 16.0 % and AUC of 91.9 % it outperforms the proposed Morphology and RSIFT methods on SE and BSE images, with respect to both metrics. The Contour RSIFT on SE images, nevertheless, outperforms the baseline with an EER of 13.1 % and AUC of 95.2 %. This result is confirmed by the receiver operator characteristics shown in Fig. 3. We see that for a fixed true positive rate, the Contour RSIFT exhibits a lower false positive rate than the baseline. The decreased performance of Contour RSIFT on BSE images is most probably caused by low textural content in the BSE images, as can be seen in Fig. 1. Comparing the EER of RSIFT and Contour RSIFT we see that just by segmenting individual particles we were able to achieve a decrease of EER by about 5 %.

TABLE I: Performance metrics (in percent) for the proposed methods and the baseline method, evaluated on the SE and BSE images.

	Metric	Method			
		Particle size	Morphology	RSIFT	Contour RSIFT
SE	EER	16.0	25.0	18.0	13.1
	AUC	91.9	84.0	89.0	95.2
BSE	EER	16.5	26.0	25.0	21.5
	AUC	92.6	84.0	85.0	90.5

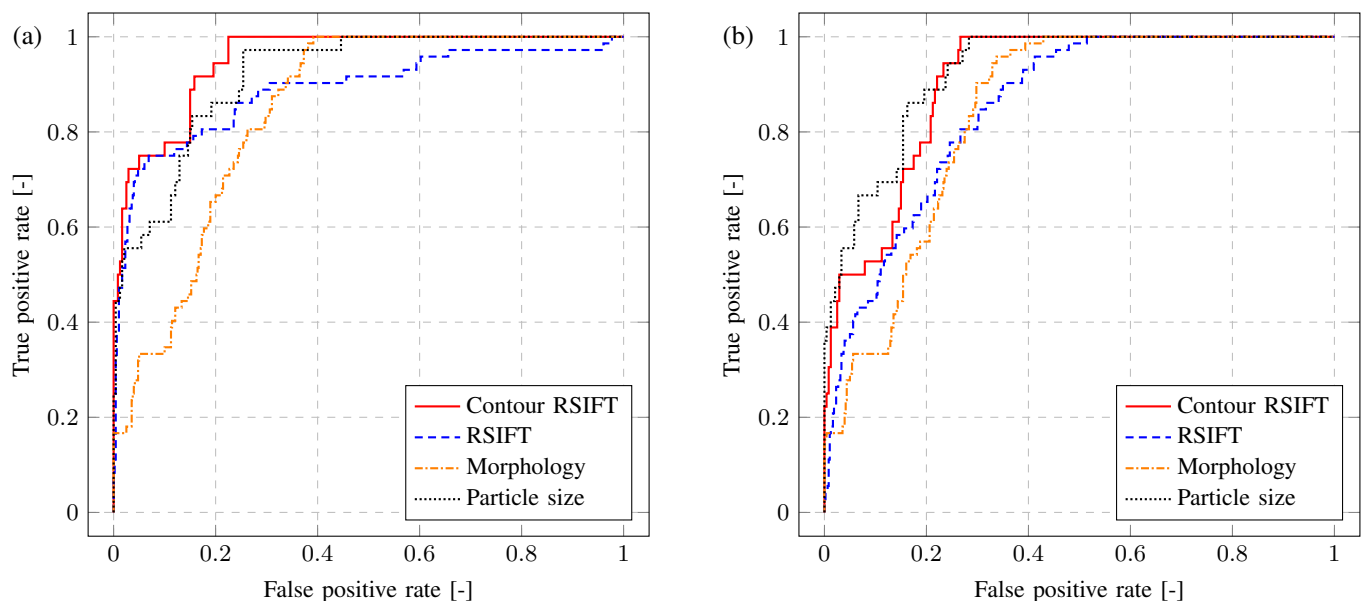


Fig. 3: Receiver operating characteristics for the proposed methods and the baseline method evaluated on (a) SE and (b) BSE images.

V. CONCLUSION

In this paper, we proposed a method for automatic comparison of soil samples, based on scanning electron microscope images of the soil particles, acquired in the SE or BSE mode. Within this method, we first extracted local descriptors from segmented or unsegmented images, which were then used to train a Gaussian mixture model (GMM). The GMM model was then used to calculate the odds (or posterior odds) that two given soil samples originate from the same location. Comparing multiple local descriptors, we found that the best performance is achieved by extracting Root SIFT descriptors along segmented contours of individual soil mineral particles. Although the achieved results are promising, we are obliged to state that this research was limited by the amount and quality of available data. In our future research we plan to evaluate our methods on a larger dataset which will be created in collaboration with the Institute of Criminalistics in Prague.

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REFERENCES

- [1] M. Kotrlý, "Možnosti analýzy stop zeminových fází v kriminalistické znalecké praxi," *Forenzní vědy, právo, kriminalistika*, no. 1, pp. 89–105, 2016.
- [2] R. C. Murray and J. C. F. Tedrow, *Forensic Geology*, Prentice Hall, Englewood Cliffs, NJ, USA, 1991.
- [3] K. Pye and S. J. Blott, "Particle size analysis of sediments, soils and related particulate materials for forensic purposes using laser granulometry," *Forensic Sci. Int.*, vol. 144, pp. 19–27, Apr. 2004.
- [4] M. Pansu and J. Gauthery, *Handbook of Soil Analysis*, Springer, New York, NJ, USA, 2006.
- [5] A. J. Newell, R. M. Morgan, L. D. Griffin, P. A. Bull, J. R. Marshall, and G. Graham, "Automated texture recognition of quartz sand grains for forensic applications," *J. Forensic Sci.*, vol. 57, no. 5, Sept. 2012.
- [6] S. Uitdehaag, W. Wiarda, T. Donders, and I. Kupier, "Forensic comparison of soil samples using nondestructive elemental analysis," *J. Forensic Sci.*, vol. 62, no. 4, July 2017.
- [7] D. Bertolini, M. L. Oliveira, E. J. R. Justino, and R. Sabourin, "Texture-based descriptors for writer identification and verification," *Expert Syst. Appl.*, vol. 40, no. 6, pp. 2069–2080, May 2013.
- [8] A. Bennour, C. Djeddi, A. Gattal, I. Siddiqi, and T. Mekhaznia, "Handwriting based writer recognition using implicit shape codebook," *Forensic Sci. Int.*, vol. 301, pp. 91–100, May 2019.
- [9] V. Christlein, D. Bernecker, F. Hönig, and E. Angelopoulou, "Writer identification and verification using GMM supervectors," in *Proc. IEEE Winter Conference on Applications of Computer Vision*, Steamboat Springs, CO, USA, Mar. 2014.
- [10] V. Christlein, D. Bernecker, F. Hönig, A. Maier, and E. Angelopoulou, "Writer identification using GMM supervectors and exemplar-SVMs," *Pattern Recognit.*, vol. 63, pp. 258–267, Mar. 2017.
- [11] N. M. Zaitoun and M. J. Mushbah, "Survey on image segmentation techniques," *Procedia Comput. Sci.*, vol. 65, pp. 797–806, 2015.
- [12] S. Minaee, Y. Boykov, F. Porikli, A. Plaza, N. Kehtarnavaz, and D. Terzopoulos, "Image segmentation using deep learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3523–3542, Feb. 2021.
- [13] A. Kirillov, E. Mintun, R. Nikhila, H. Mao, C. Rolland, L. Gustafson, T. Xiao, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, "Segment anything," Apr. 2023.
- [14] C. Christophe, A. Biedermann, J. Vuille, S. Willis, and J. De Kinder, "ENFSI guideline for evaluative reporting in forensic science: A primer for legal practitioners," *Criminal Law and Justice Weekly*, vol. 180, no. 10, pp. 189–193, Mar. 2016.
- [15] G. Bradski, "The OpenCV library," *Dr. Dobbs's Journal of Software Tools*, 2000.
- [16] R. Arandjelović and A. Zisserman, "Three things everyone should know to improve object retrieval," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Providence, RI, USA, June 2012.
- [17] D. G. Tzikas, A. C. Likas, and N. P. Galatsanos, "The variational approximation for Bayesian inference," *IEEE Signal Process. Mag.*, vol. 25, no. 6, pp. 131–146, Nov. 2008.
- [18] D. A. Reynolds, T. F. Quatieri, and R. B. Dunn, "Speaker verification using adapted Gaussian mixture models," *Digital Signal Process.*, vol. 10, no. 1-3, pp. 19–41, Jan. 2000.
- [19] S. P. Lund and H. Iyer, "Likelihood ratio as weight of forensic evidence: A closer look," *J. Res. Natl. Inst. Stan.*, vol. 122, no. 27, Oct. 2017.
- [20] D. G. Lowe, "Object recognition from local scale-invariant feature," in *Proc. Int. Conf. Comput. Vis.*, Kerkyra, Greece, Sept. 1999.
- [21] N. Kroell, "Imea: A python package for extracting 2D and 3D shape measurements from images," *J. Open Source Softw.*, vol. 6, no. 60, Apr. 2021.